Listing Outcomes, Tables and Tree Diagrams

Name _____ Date _____ No calculator Version 1

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KEY WORDS outcome *(putanga)* · equally likely *(tūpono ōrite)* · tree diagram *(hoahoa rākau)*

A Listing outcomes ●○○

Remember: Be systematic: keep the first choice fixed and go through every second choice, then move on to the next first choice.

1. List all the possible outcomes. Use the letters and numbers given.

a) A coin is tossed twice (H, T).

b) A coin is tossed (H, T) and a spinner numbered 1, 2, 3 is spun.

c) You choose a sandwich (cheese C, ham H, egg E) and a drink (juice J, water W).

d) A spinner with the letters A, B, C is spun twice.

B Outcomes in a table ●●○

Remember: A table shows every outcome of two events once. Each cell is one outcome, and every cell is equally likely.

2. A spinner numbered 1 to 4 and a spinner numbered 1 to 3 are spun, and the two numbers are added. Complete the table to show every possible total.

+	1	2	3	4
1				
2				
3				

3. Use your table.

a) How many outcomes are there altogether? _____

b) What is the largest possible total? _____

c) How many outcomes give a total of 5? _____

d) How many outcomes give a total of less than 4? _____

4. Find the probability that the total is:

a) 5 _____ **b)** 2 _____

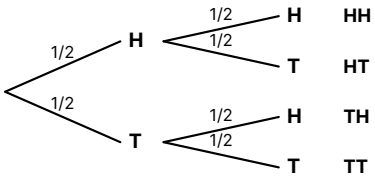
c) an even number _____ **d)** more than 5 _____

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C Tree diagrams ●●●

> **Remember:** Each path along the branches of a tree diagram is one outcome. Follow a branch from the first stage, then a branch from the second stage.

5. The tree diagram shows a coin being tossed twice.



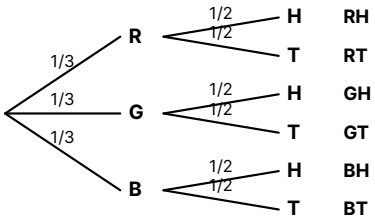
a) How many outcomes does the tree diagram show?

b) Find the probability of getting two heads.

c) Find the probability of getting one head and one tail, in either order.

d) Find the probability of getting at least one tail.

6. A spinner has three equal sections: red, green and blue (R, G, B). The spinner is spun and then a coin is tossed.



a) How many outcomes does the tree diagram show?

b) Find the probability of getting red and a head.

c) Find the probability of getting a tail.

d) Find the probability of getting a colour that is not blue, and a head.

D Counting outcomes ●●○

> **Remember:** When there are two stages, multiply the number of choices at each stage. A coin (2) and a dice (6) give $2 \times 6 = 12$ outcomes.

7. How many outcomes are there?

a) A coin is tossed twice _____

b) A coin is tossed and a dice is rolled _____

c) 2 shirts and 3 pairs of shorts _____

d) 4 sandwiches and 3 drinks _____

e) Two dice are rolled _____

f) A coin is tossed three times _____

★ Extension

8. A coin is tossed three times. List the 8 outcomes first, then find the probability of getting:

a) three heads

b) exactly two heads

c) at least one head

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Listing Outcomes, Tables and Tree Diagrams

Name _____ Date _____ No calculator Version 2

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KEY WORDS outcome *(putanga)* · equally likely *(tūpono ōrite)* · tree diagram *(hoahoa rākau)*

A Listing outcomes ●○○

Remember: Be systematic: keep the first choice fixed and go through every second choice, then move on to the next first choice.

1. List all the possible outcomes. Use the letters and numbers given.

a) A coin is tossed twice (H, T).

b) A coin is tossed (H, T) and a spinner numbered 1, 2, 3, 4 is spun.

c) You choose an ice cream (vanilla V, chocolate C, mint M) and a topping (nuts N, sauce S).

d) A spinner with the letters X, Y, Z is spun twice.

B Outcomes in a table ●●○

Remember: A table shows every outcome of two events once. Each cell is one outcome, and every cell is equally likely.

2. A spinner numbered 1 to 4 and a spinner numbered 1 to 4 are spun, and the two numbers are added. Complete the table to show every possible total.

+	1	2	3	4
1				
2				
3				
4				

3. Use your table.

a) How many outcomes are there altogether? _____

b) What is the largest possible total? _____

c) How many outcomes give a total of 5? _____

d) How many outcomes give a total of less than 4? _____

4. Find the probability that the total is:

a) 5 _____ **b)** 2 _____

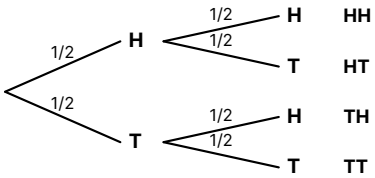
c) an even number _____ **d)** more than 5 _____

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C Tree diagrams ●●●

Remember: Each path along the branches of a tree diagram is one outcome. Follow a branch from the first stage, then a branch from the second stage.

5. The tree diagram shows a coin being tossed twice.



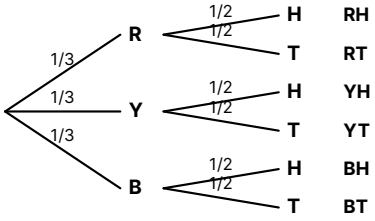
a) How many outcomes does the tree diagram show?

b) Find the probability of getting two heads.

c) Find the probability of getting one head and one tail, in either order.

d) Find the probability of getting at least one tail.

6. A spinner has three equal sections: red, yellow and blue (R, Y, B). The spinner is spun and then a coin is tossed.



a) How many outcomes does the tree diagram show?

b) Find the probability of getting red and a head.

c) Find the probability of getting a tail.

d) Find the probability of getting a colour that is not blue, and a head.

D Counting outcomes ●●○

Remember: When there are two stages, multiply the number of choices at each stage. A coin (2) and a dice (6) give $2 \times 6 = 12$ outcomes.

7. How many outcomes are there?

a) A coin is tossed twice _____

b) A coin is tossed and a dice is rolled _____

c) 3 shirts and 2 pairs of shorts _____

d) 5 sandwiches and 3 drinks _____

e) Two dice are rolled _____

f) A coin is tossed three times _____

★ Extension

8. A coin is tossed three times. List the 8 outcomes first, then find the probability of getting:

a) three heads

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